

Fitting Exponential Family Distributions to Observed Stomach Contents Data for Certain Species of Fish

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Abstract

Size spectrum models are a key tool in the arsenal of fisheries scientists. A central modelling assumption for such models is the feeding kernel which expresses the preferred prey size of predator species. Hence, analytic expressions for the feeding kernel may be useful in the further development of size spectrum models.

This report theoretically links the observed stomach contents data to the feeding kernel by means of the Sheldon spectrum and introduces some key statistical concepts, in particular the exponential family of distributions and a multiplicative property of Gamma distributed random variables. In the data section, first the statistics of interest are introduced and justification is made for the time and geographic refinement of data. Then the report comprehensively analyses the observed predator-prey mass ratio of six chosen species (cod, haddock, herring, plaice, sole and whiting) and members of the exponential family of distributions are fitted to these data. The report concludes with the calculation of the log likelihood on a species-wise basis to quantify how well the fitted distributions model the observed stomach contents data for the species at hand.

1 Introduction

The responsible, sustainable management of fisheries is a challenge for governments in many countries. For example, a stated aim of the European Union's Common Fisheries Policy (CFP) is to "ensure sustainable fisheries". [1]

Almost all nations that regulate the sea do so on the basis of scientific advice [Hutchings, 2022]. [2] [3] One tool scientists use to analyse what a sustainable exploitation of the fishery is are size spectrum models (for example Mizer [4]).

A key parameter in such size spectrum models is the feeding kernel of fish — effectively what size prey do predators prefer to eat in the ocean. This report first deals with the feeding kernel and its relationship to the size of prey observed in the stomach and why having analytic expressions for the feeding kernel may be useful in the continued development of size spectrum models. Next, certain statistical tools are introduced, namely kernel density estimation and the exponential family of distributions. Certain results about the Gamma distribution are formulated in terms of estimating its parameters and its multiplicative properties. Finally data are analysed to ascertain the suitability of fitting exponential family distributions to observed data from fish stomachs and the log likelihood is calculated for the fitted curves as a means of quantifying the appropriateness of the Gamma estimate.

2 Feeding Preference and Practice

We must be careful to distinguish between three concepts that are distinct but related, namely the feeding kernel, abundance and realised stomach contents.

The predator-prey mass ratio is defined in an intuitive way and is the ratio of the mass of the predator to the mass of a single consumed item of prey.

The first concept is that of the feeding kernel — simply what size of prey do predator species prefer to eat. For example, it may be known that a particular species has a preference for eating fish that have a mass 10^{-3} times the mass of itself. An upper bound on the size of prey a predator can eat is given by physiological considerations such as the size of the fish's mouth. On the other hand, a lower bound on preferred prey mass could be caused by a trade-off between effort to catch the prey and the amount of biomass gained by it.

[1] The common fisheries policy: origins and development, European Parliament, Date Accessed = 18/4/22, https://www.europarl.europa.eu/factsheets/en/sheet/114/the-common-fisheries-policy-origins-and-development.

[2] Ibid.

[3] Sustainable fisheries for future generations, UK Government, Date Accessed = 25/4/22, https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/722074/fisheries-wp-consult-document.pdf.

[4] https://sizespectrum.org/mizer/, Date Accessed = 25/4/22.

The next concept is that of the abundance. Simply, a fish cannot eat prey of a certain size if that prey doesn't exist in the part of the ocean the fish is in. The abundance of prey can be modelled by the Sheldon spectrum as explained in the next subsection and in general is measured as the number of potential prey items of a certain size.

Finally, the realised stomach contents is what can actually be observed in the stomach. This measures what the fish has been able to successfully predate on. Suppose that the we have an expression for the feeding kernel $\varphi_{w'}(w)$ as a function of the prey weight and an expression for the prey abundance $N(w)$. Let $I_{w'}(w)$ be the number of prey items of weight w in the stomach of a predator of weight w'. Then the realised stomach contents is proportional to the product of the feeding kernel and the abundance so that

$$I_{w'}(w) \propto \varphi_{w'}(w)N(w). \tag{1}$$

2.1 Sheldon Spectrum

The question of approximating the abundance of prey is made much simpler by the Sheldon spectrum [Sheldon et al., 1972].

The Sheldon Spectrum states that there is a "roughly equal concentration of material in all size grades". This is with respect to bulk, ie. mass. Perhaps more instructively, in an example from the paper,

> given a distribution between the limits of 1 μ and 1mm... a third [of the bulk] will lie in each of 1-10μ, 10-100μ and 100μ to 1mm.

On the basis of the Sheldon Spectrum, it is possible to approximate the amount of biomass between given sizes in the ocean and thence to approximate the number of individuals of a given size.

A humorous application of his eponymous spectrum was made by R.W. Sheldon in [Sheldon and Kerr, 1972] where he argues that on the basis of the fish yield in Loch Ness, there are 10 to 20 monsters in Loch Ness with weight around 1500kg.

Now, the Sheldon Spectrum is, in effect, a power law. It is understood that all power laws do eventually break down. It is therefore reasonable to think there be 1,000,000,000 individuals of mass 1g and 1,000,000 individuals of mass 1kg in a sea. However, it is understood that there is not one individual of mass 1,000,000kg in the sea.

We consider the distribution of mass of fish to be a continuous random variable. Therefore, let the density of the number of fish of a given weight be $N(w)$.

The Sheldon spectrum implies that

$$N(w) = nw^{-2}. \tag{2}$$

2.2 Usefulness of Stomach Data

In this report, we assume that stomach data provides a reasonable estimate for the realised catch of prey. For simplicity, we ignore elements such as digestion and satiation. Then by using the Sheldon spectrum as an estimate for the abundance, we can divide the observed stomach contents by abundance which is a power law to obtain an approximation to the feeding kernel of various species.

Now, if it is possible to use a member of the exponential family of distributions to estimate the realised catch and since the abundance is of the form $\exp tw$ where w is prey weight then it may be possible to use an exponential family distribution for the feeding kernel. This would give an analytic expression for the feeding kernel that could then be used in size-spectrum models such as Mizer.

Stomach data is used for other puropses as well in reasearch. For example [Johnson et al., 2013] uses stomach data to show that "cod predation was by far the most significant predator of *Nephrops [norvegicus]*" in the Irish Sea. Note that *Nephrops [norvegicus]* is the binomial name for Norway lobster or Dublin Bay prawns.

3 Kernel Density Estimation

Kernel Density Estimation (KDE) will be an important tool used later in this dissertation and therefore it has been deemed expedient to include a brief summary of it here.

Suppose we have n independent, identically distributed random variables $X_1, X_2, \ldots, X_n$ (known as a sample) from a distribution whose density function, $f_X(x)$ is not known but is assumed to exist. Then Kernel Density Estimation (also known as Parzen Windows or the empirical probability density function) provides an easy, if computationally intensive [Dekking et al., 2005], way of estimating the density function and, as shown in [Parzen, 1962], will converge to the true distribution as the size of the sample n tends to infinity.

3.1 Historical context

Kernel Density Estimation (KDE) was only seriously considered in research beginning in the late 1950's. At its core, KDE seeks to infer the nature of the probability density function (pdf) of a random variable given a sample from that random variable. Prior to the development of KDE by [Rosenblatt, 1956] & [Parzen, 1962] in the mid 20th century, inference of pdfs from data was limited to histograms. Histograms are of limited usefulness because they are very discrete and a small change in bin width or shift in bin location may result in a plot of a very different nature [Dekking et al., 2005]. The lack of research was in spite of the admission by [Parzen, 1962] of "the obvious importance of these [estimation] problems."

3.2 The Method

Intuitively, one expects areas with a high density of observations to have a high value of the density function, and those areas that are more sparse to have a lesser density function.

To that end, a kernel function $K(y)$ is introduced which in general is a probability density, is radially symmetric and unimodal [Dekking et al., 2005]. The kernel function is evaluated at each data point and then all these functions are summed together, with an appropriate normalisation constant (namely n^{-1} where n is the number of samples) to give the empirical probability density function. That is to say, given a sample $(X_i)_{i=1}^n$ and assuming a bandwidth h, the empirical probability density function, $\hat{f}(x)$ is [Sheather and Jones, 1991]

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K\left(x - x_i\right). \tag{3}$$

3.3 Choices and parameters

Fundamentally, there are two key choices to be made when implementing KDE, viz. the choice of kernel function and the choice of bandwidth. We consider both in turn.

3.3.1 Choice of Kernel Function

The two most common choices of kernel function are the rectangular and the Gaussian, although others exist such as Epanechnikov [Epanechnikov, 1969] and the triangular kernel.

The simplest choice of kernel function is the rectangular (or window) function which has the formula:

$$K_r(x) = \begin{cases} \frac{1}{h} & \text{if } |x - x_i| \leq \frac{h}{2}, \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

and can be seen in Figure 1 on the left. This kernel function has the disadvantage of weighing all points within a certain distance (the bandwidth) of the data point equally. It would be natural to assume that we should weigh points closest to the data point the heaviest.

The Gaussian kernel function is precisely that, each data point contributes a Gaussian probability and the empirical density function is the sum of these Gaussians. The formula for this type of density function is

given (in 1-dimension) by

$$K_G(x) = \frac{1}{h\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x - x_i}{h}\right)^2\right). \tag{5}$$

The shape of this kernel function is the familiar bell-curve as can be seen on the right hand side of Figure 1.

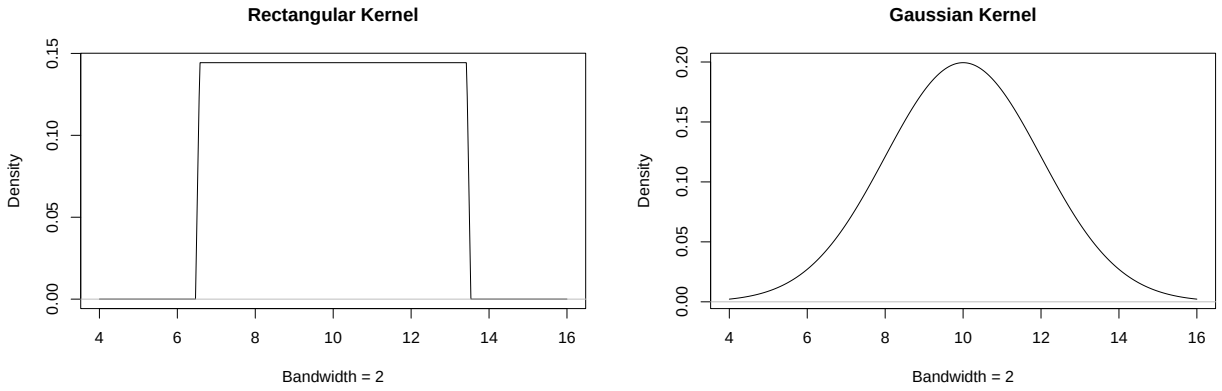


Figure 1: The window and Gaussian Kernel Functions, both centred at 10 with bandwidth $h = 2$.

3.3.2 Choice of Bandwidth

The bandwidth (h in expressions above) is the breadth of influence each data point has. Its selection is always a trade off between being overly specific (with a narrow bandwidth) and hiding the underlying nature of the data (with too wide a bandwidth). R, by default, will use the method described in [Sheather and Jones, 1991] to choose the value for h.

3.4 Example of KDE with a Small Sample from a Normal Distribution

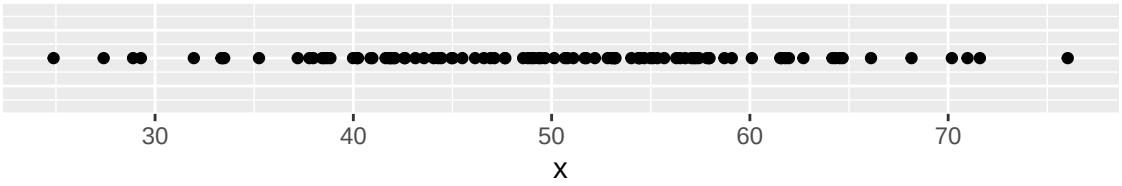


Figure 2: 100 samples taken from a $N(50, 10)$ distribution.

We first demonstrate KDE in practice and the importance of the choice of kernel function by considering a random variable $X \sim N(50, 10)$ of which we take 100 i.i.d. samples, namely $X_1, X_2, \ldots, X_{100}$ and plot in Figure 2.

The data points are more concentrated towards the mean, as expected. We now plot the kernel density estimate using both the window and Gaussian kernel functions, and include the true density function (dashed line) since $X \sim N(50, 10)$.

The Gaussian kernel in this case is more appropriate since gives a much smoother empirical pdf with an unlimited support like the distribution that has been sampled from, however it is more computationally expensive.

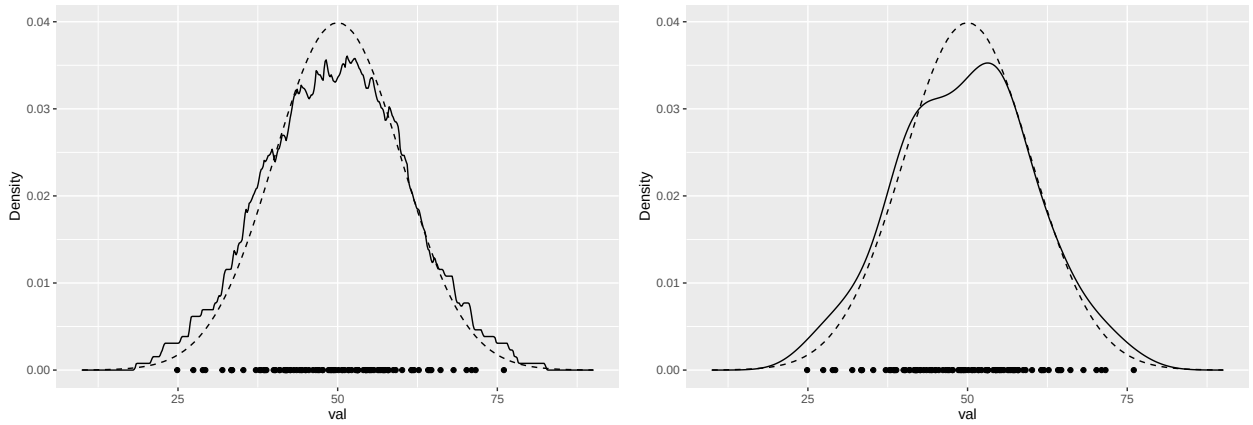


Figure 3: Evaluating the emprirical probability density using the window (left) and Gaussian (right) functions with default bandwidth.

4 The Exponential Family of Distributions

The exponential family of distributions is a broad family of distributions where the probability density has the general form

$$f(x|\eta) = h(x) \exp\left(\eta^\top T(x) - A(\eta)\right) \tag{6}$$

where η is a parameter vector, known as the *canonical parameter*, $T(x)$, $h(x)$ are functions and $A(\eta)$ is named the *cumulant function* [Jordan, 2010].

We look at some specific examples.

4.1 Exponential Distribution

The exponential distribution is the simplest exponential family distribution and simply has an exponential decay of the probability density as x increases. The distribution has probability density function

$$f(x|\lambda) = \begin{cases} 0 & \text{if } x < 0 \\ \exp(\lambda x) & \text{if } x \geq 0. \end{cases}$$

In terms of the formulation in Equation 6, $\eta = \lambda$, $T(x) = x$, $A(\eta) = 0$ and

$$h(x) = \mathbb{1}_+(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } x \geq 0. \end{cases}$$

The exponential distribution is very flexible and can be used to model chemical reactions, stopping times and reliability. This distribution will not be used extensively in this project but is included for completeness with respect to the distribution family.

4.2 Normal Distribution

The Normal distribution is one of the most central in probability theory and statistics [Dekking et al., 2005] due to the frequency it is observed in many contexts (as diverse as psychology, engineering and trading standards). The Normal distribution is also fundamental to the Central Limit Theorem.

The familiar Gaussian bell curve is not reproduced here but can be seen on the right hand side of Figure 3. The probability density function of the standard Normal distribution $Z \sim N(0, 1)$ is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right). \tag{7}$$

For a general Normal random variable with mean μ and variance σ^2, it can be written in general exponential family form (taken from [Jordan, 2010]) as

$$\eta = \begin{bmatrix} \mu/\sigma^2 \\ -1/2\sigma^2 \end{bmatrix} \tag{8}$$

$$T(x) = \begin{bmatrix} x \\ x^2 \end{bmatrix} \tag{9}$$

$$A(\eta) = \frac{\mu^2}{2\sigma^2} + \log \sigma \tag{10}$$

$$h(x) = \frac{1}{\sqrt{2\pi}}. \tag{11}$$

4.3 Log-normal Distribution

The log-normal distribution is a probability distribution whose logarithm is normally distributed. It is particularly important in the context of size-spectrum models as Mizer assumes the predation kernel is log-normally distributed [Scott et al., 2014].

The log-normal is related to the Normal distribution by the relationship that if $X \sim \text{Lognormal}(\mu_x, \sigma_x^2)$ then $\log X \sim N(\mu, \sigma^2)$. This distribution X has the probability density function

$$f_X(x|\mu, \sigma^2) = \frac{1}{x\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right). \tag{12}$$

4.4 Gamma Distribution

The Gamma distribution will play a key role in later sections of this dissertation. It is parametrised by two parameters — namely the shape α and the rate β. Then a Gamma distributed random variable $W \sim \text{Gamma}(\alpha, \beta)$ has probability density function

$$f_W(w|\alpha, \beta) = f^{\Gamma}_{\alpha,\beta}(w) = \frac{\beta^\alpha}{\Gamma(\alpha)} w^{\alpha-1} e^{-\beta w}. \tag{13}$$

In the terms of the formulation of exponential distributions as given in Equation 6,

$$\eta = \begin{bmatrix} \alpha - 1 \\ -\beta \end{bmatrix} \tag{14}$$

$$T(w) = \begin{bmatrix} \log w \\ w \end{bmatrix} \tag{15}$$

$$A(\eta) = 0 \tag{16}$$

$$h(w) = \frac{\beta^\alpha}{\Gamma(\alpha)}. \tag{17}$$

4.5 Estimating Parameters of Gamma Distributed Random Variables

In later sections, estimation of parameters of Gamma distributed random variables will be imperative. The method of moments provides a computationally simple way of estimating the parameters of the Gamma distribution in terms of its sample mean and variance by correctly matching the sample mean and variance to the parameters.

Given a Gamma distributed random variable $W \sim \text{Gamma}(\alpha, \beta)$ we can estimate the parameters as $\hat{\alpha}$ and $\hat{\beta}$ using the method of moments estimator as

$$\hat{\alpha} = \frac{\overline{W}^2}{\hat{\sigma}^2_W} \;\&\; \hat{\beta} = \frac{\overline{W}}{\hat{\sigma}^2_W}. \tag{18}$$

Where $\overline{W}$ is the sample mean and $\hat{\sigma}^2_W$ is the sample variance.

4.6 Multiplicative Properties of Exponential Family Distributions

The reason we express such an interest in the exponential family of distributions is that when multiplied together, they maintain similar statistical features.

As a consequence of the Sheldon spectrum, if we have fitted an exponential distribution to the observed data, we may want to multiply this by a factor of $e^{\lambda x}$.

It is well known that when the exponential or Normal distributions are multiplied in this way, they remain as the same distribution and the parameters shift a predictable amount.

Now we consider whether this is true for the Gamma distribution.

Suppose $W \sim \text{Gamma}(\alpha, \beta)$. Then

$$e^{\lambda w} f^{\Gamma}_{\alpha,\beta}(w) = e^{\lambda w} \frac{\beta^{\alpha}}{\Gamma(\alpha)} w^{\alpha-1} e^{-\beta w} \tag{19}$$

$$= \frac{\beta^{\alpha}}{\Gamma(\alpha)} w^{\alpha-1} e^{-(\beta-\lambda)w}. \tag{20}$$

Now let $c = \beta - \lambda$ so

$$e^{\lambda w} f^{\Gamma}_{\alpha,\beta}(w) = \frac{(c+\lambda)^{\alpha}}{\Gamma(\alpha)} w^{\alpha-1} e^{-cw} \tag{21}$$

$$= \frac{(c+\lambda)^{\alpha}}{c^{\alpha}} \frac{c^{\alpha}}{\Gamma(\alpha)} w^{\alpha-1} e^{-cw} \tag{22}$$

$$e^{\lambda w} f^{\Gamma}_{\alpha,\beta}(w) = \frac{(c+\lambda)^{\alpha}}{c^{\alpha}} f^{\Gamma}_{\alpha,c}(w) \tag{23}$$

$$e^{\lambda w} f^{\Gamma}_{\alpha,\beta}(w) = \frac{\beta^{\alpha}}{(\beta-\lambda)^{\alpha}} f^{\Gamma}_{\alpha,\beta-\lambda}(w). \tag{24}$$

In consequence, when a $\text{Gamma}(\alpha, \beta)$ random variable is multiplied by $e^{\lambda w}$, it results in a function that is proportional to the probability density function of a $\text{Gamma}(\alpha, \beta - \lambda)$ random variable. To ensure that the pdf integrates to 1, we may consider multiplying $W \sim \text{Gamma}(\alpha, \beta)$ by $\frac{(\beta-\lambda)^{\alpha}}{\beta^{\alpha}} e^{\lambda w}$ which does then make a new random variable distributed according to $\text{Gamma}(\alpha, \beta - \lambda)$.

5 Data Analysis Methods

In this second section of the dissertation, we turn our attention to ascertain whether it is possible to fit any of the exponential family of distributions to data observed.

5.1 Data Used

The data used for this reports was provided by the Centre for Environment, Fisheries and Aquaculture Science (CEFAS) and is an amalgamation of DAPSTOM and ICES data. The DAPSTOM data consists of 110243 observations and is described in [Pinnegar, 2019]. The data from the International Council for the Exploration of the Sea (ICES) consists of 190295 observations and is described in [ICES, 2010].

This provides data over a very long timeframe ranging from 1836 to 2016. In our analysis, we will consider whether changes in the seas obscure some of the trends in the data.

5.2 General Overview of Data

Both datasets have been formed using stomach observations.

Put simply, researchers catch fish in the sea, cut them open and observe what is found within their stomachs. The larger fish (the predator) and the stomach contents are weighed and the species recorded. In many cases,

other observations are made such as the length of the predator and prey. Occasionally, it may be necessary to bin stomach contents in order to find prey, but where this has been done, appropriate corrections have been applied to the data to ensure results accurately reflect what was found on an individual basis.

5.3 Statistics

It is worthwhile to set out what we are looking for in the data.

One of the main statistics of interest is the predator-prey mass ratio [Reum et al., 2019b] which is (quite intuitively) defined as the ratio of the predator mass to the prey mass. The feeding kernel is a function of the predator-prey mass ratio and so this is clearly a central statistic in our analysis.

In order to visualise the data and measure the closeness of the fitted curves to the actual distribution we will make extensive use of the technique of Kernel Density Estimation (KDE), an overview of which is provided in a separate chapter.

In general, we will with work with log statistics as is both common in literature (eg [Reum et al., 2019a]) and is consistent with the discussion in the exponential chapter. For example, Mizer assumes the feeding kernel is log-normal. This means that for a statistic X, we will in general work with the variable

$$\tilde{X} = \log \frac{X}{1_U} \tag{25}$$

where 1_U is the unit in which X is measured. (This is necessary because, for example $\log 13$ mm does not make sense.)

5.4 Aims

We aim to see if it is possible to approximate empirical data with standard probability distributions.

In particular, we will consider statistics such as the predator prey mass ratio and, by applying Kernel Density Estimation observe whether the empirical probability distribution is reminiscent of any well-known probability density functions. If the empirical probability density function is deemed satisfactorily close to a well-known probability density then parameters of such a distribution will be estimated and the appropriateness of the estimate quantified by calculating the log likelihood for the data.

5.5 Initial Observation

As has been previously justified, we plot the log PPMR.

The species chosen to focus on are cod (*Gadus morhua*), haddock (*Melanogrammus aeglefinus*), herring (*Clupea harengus*), plaice (*Pleuronectes platessa*), sole (*Solea solea*) and whiting (*Merlangius merlangus*). Log PPMR plots for these species are found in Figure 4.

On initial observation, we see a wide variety of curve shapes which suggests that it won't initially be possible to fit one distribution to all the species. However, most of the species (apart from herring) do have a peak of density in the region of 5 to 10 meaning that, in general, fish tend to eat prey that are $e^5 \approx 148$ to $e^{10} \approx 22000$ times smaller than themselves.

The case with whiting lends itself to a Gamma distribution and we will try fitting parameters to this. For the other species, we will refine the data on the basis of time or geography to try and obtain a reasonably fitting probability distribution.

5.6 Time and Geographic Analysis

It is worth justifying here why breaking up the data in terms of time and space is reasonable.

The world's seas are diverse ecosystems with different species making their homes in different areas. In particular, some species are more suited to the cold seas north of Norway and Russia whilst other species

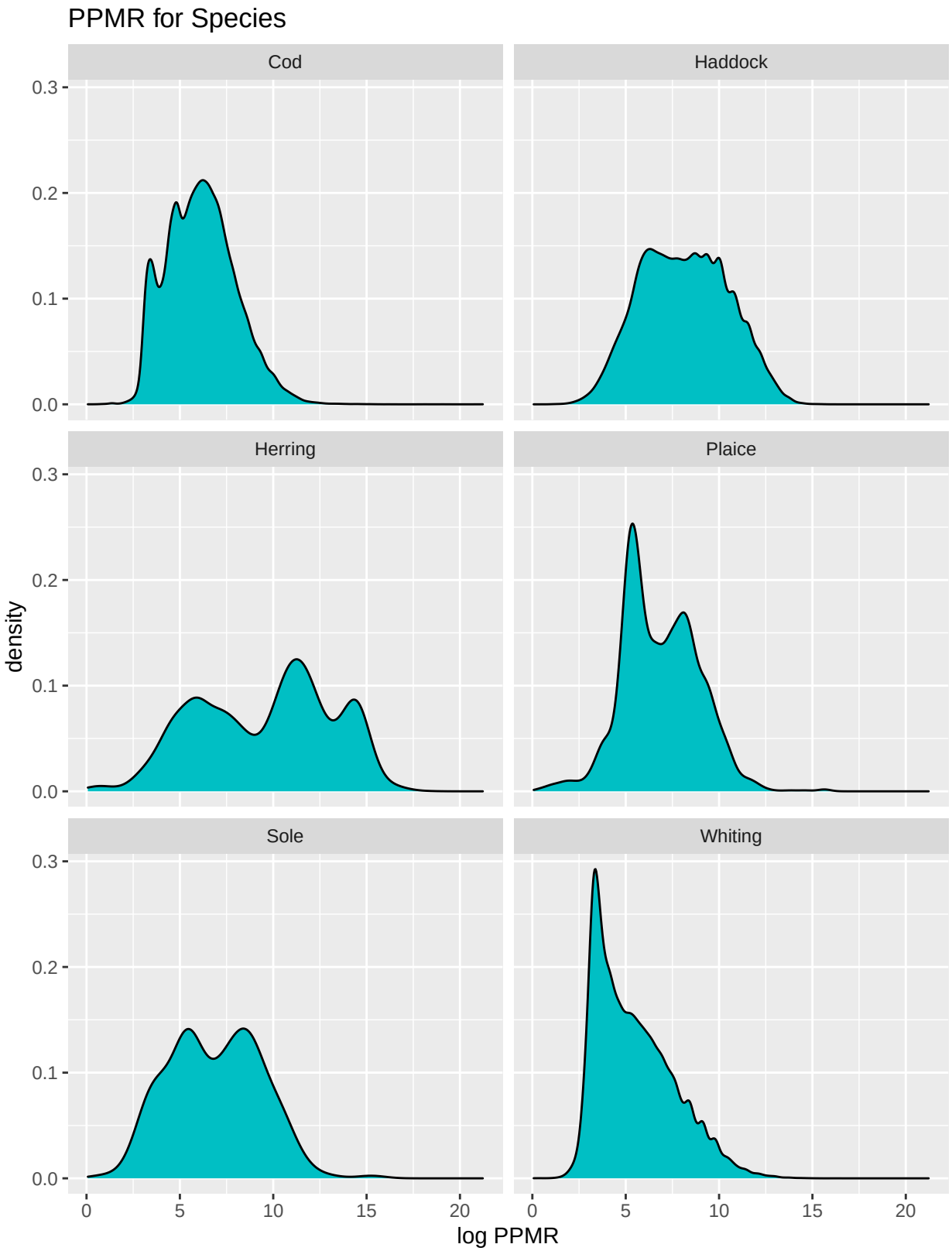


Figure 4: Log PPMR plots for our chosen species.

prefer the relatively warmer waters of the Celtic Sea. Note that we restrict ourselves to seas in north west Europe as this is the geographic extent of the stomach data. As a result of the large diversity of species that can be predated upon, it is reasonable to restrict ourselves to seas when analysing the data as different seas may have different abundances of prey. If we don't allow ourselves this flexibility, we may find that it is impossible to appropriately fit a distribution to the data.

The stomach dataset provided by Cefas ranges over 180 years. Understandably, there have been significant changes over that time. Of prime concern is industrialisation which has enabled more intensive exploitation of fisheries. There have also been significant policy changes over this time. At the beginning of the period, fishing was a manpower heavy industry but was reminiscent of today's industry with steam powered trawlers able to cover long distances.

The Second World War (1939—1945) was a significant disruptor to fishing since many fishing vessels were requisitioned by national Admiralties. In the aftermath of the war, the significant technological advances caused by conflict found their way onto trawlers. According to [Holm, 2012],

> Post-war trawlers were not only faster and possessed greater fishing capacity than their predecessors, they were also equipped with all the positioning, fish-finding and communication technology developed during the war, which helped them find and catch more fish.

Prior to the Second World War, there was open access to all ocean resources with certain exceptions such as the North Sea Fisheries Convention (1882). Many nations following the war exploited fisheries heavily without paying due regard to the conservation of fisheries stock [Holm, 2012]. Consequently, a number of cataclysmic collapses in fish stocks have been observed. For example there was a collapse in North Sea herring stock in 1975 and a collapse in the north west Atlantic cod fishery in 1992. As the Second World War did play such a large part in the history of fisheries, I consider the 'post-war' period to start in 1950 (time was needed, for example to return requisitioned vessels). Therefore, as a general time splitting point, I will take 1950. However, in light of the North Sea collapse in herring populations in 1975, that year will be applied to herring.

5.7 Challenges

It may be worthwhile noting that due to the very nature of kernel density estimation 'spreading out' the probability density, there are some $x < 0$ such that $f(x) > 0$ which would mean that the predator was eating a fish bigger than itself which is unlikely. (Although some dogfish do take bites out of their prey.)

5.8 Quantifying Closeness of a Fitted PDF

There are various methods that could be used to quantify how well the fitted probability density fits the observed data. Our method of choice will be to use the likelihood.

Suppose that $x_1, x_2, \ldots, x_n$ are observations from a sample $X_1, X_2, \ldots, X_n$ that we assume is independent and identically distributed. Then the likelihood, $L(\vec{\theta})$ given a parameter vector $\vec{\theta}$ is

$$L(\vec{\theta}|x_1, x_2, \ldots, x_n) = f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n) \tag{26}$$

where $f_{X_1,X_2,\ldots,X_n}$ is the joint probability density function.

Now since we assume that $X_1, X_2, \ldots, X_n$ are independent and identically distributed, we can factorise the right hand side of Equation 26 giving

$$L(\vec{\theta}) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n). \tag{27}$$

Now we define the log-likelihood $\ell(\vec{\theta})$ as

$$\ell(\vec{\theta}) = \log L(\vec{\theta}) \tag{28}$$

then use the multiplicative property of the logarithm and that the random variables are identically distributed to give

$$\ell(\vec{\theta}) = f(x_1|\vec{\theta}) + f(x_2|\vec{\theta}) + \cdots + f(x_n|\vec{\theta}) = \sum_{i=1}^{n} f(x_i|\vec{\theta}). \tag{29}$$

In general, we will use $\ell(\vec{\theta})/n$ where n is the number of observations as this will allow us to quantitatively compare the closeness of fit across species.

6 Data Results

The results from the various species are presented on a species-wise basis including details as to whether they have been refined with respect to time or geography.

6.1 Whiting

Upon initial inspection of Figure 4 the whiting data seems similar to a Gamma distribution.

Applying this to the data at hand, we find $\hat{\alpha} = 6.83$ and $\hat{\beta} = 1.25$ using the method of moments estimator as described above.

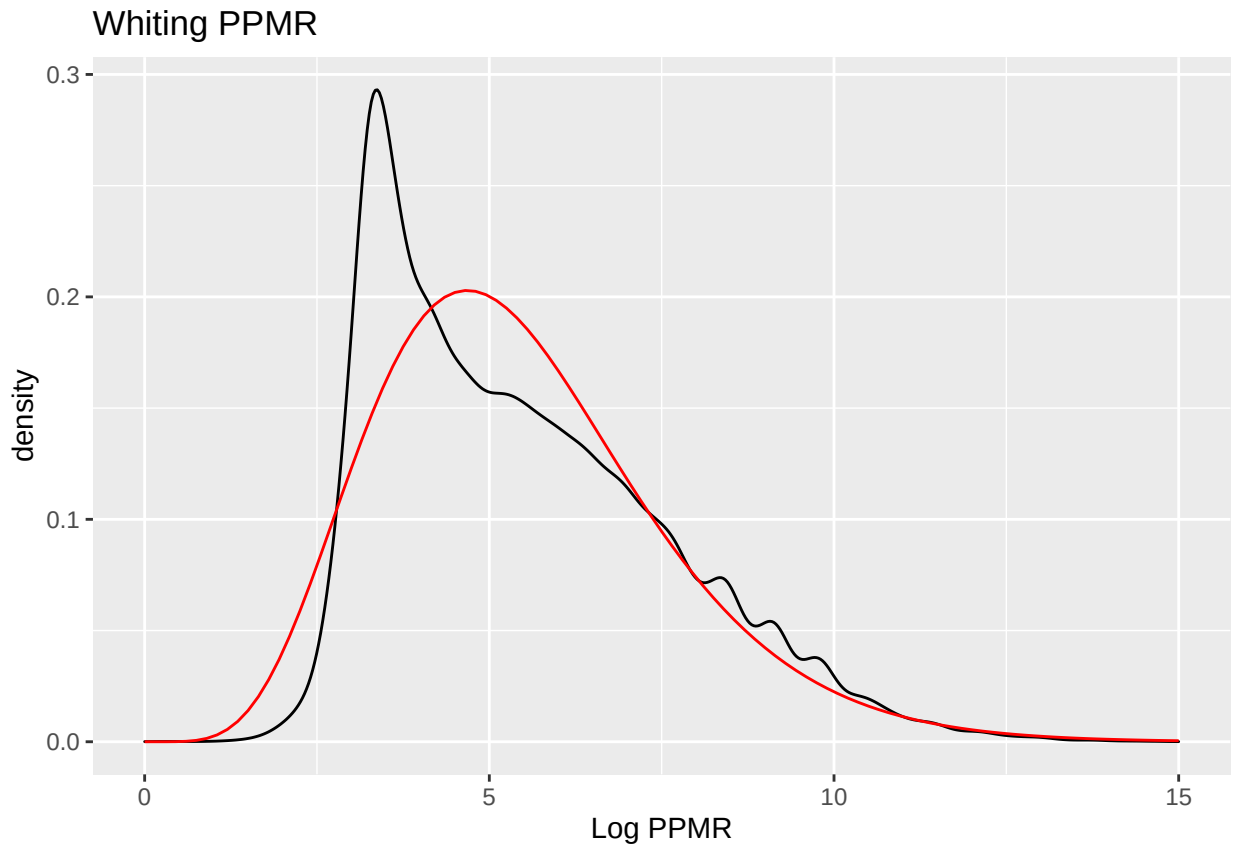


Figure 5: Whiting PPMR data with estimated Gamma Distribution.

Plotting the Gamma distribution with these estimated parameters gives Figure 5. The fitted Gamma distribution fits well- especially at either tail but it does not capture the extremity of the peak at the mode of the density estimate.

We can calculate the log-likelihood of this estimate as described above giving $\ell(\hat{\alpha}, \hat{\beta}) = -1.4731211 \times 10^5$ which corresponds to a log-likelihood of -2.0833574 per observation.

6.2 Plaice

Recalling Figure 4, the plaice does seem bimodal and so we consider whether it is possible to refine the data to make a unimodal distribution that fits a standard probability distribution.

In the first instance, we refine on a time basis. By the justification given above for splitting in 1950, we consider the pre- and post-1950 data. There are 2092 observations pre-1950 and 2406 observations post-1950 which is a reasonable amount of data to make inferences on.

Now the post-1950 log PPMR plot as shown in the right hand side of Figure 6 keeps, and amplifies the bimodality evident in Figure 4. This suggests that it will not be possible to fit a standard probability distribution to the post-1950 plaice.

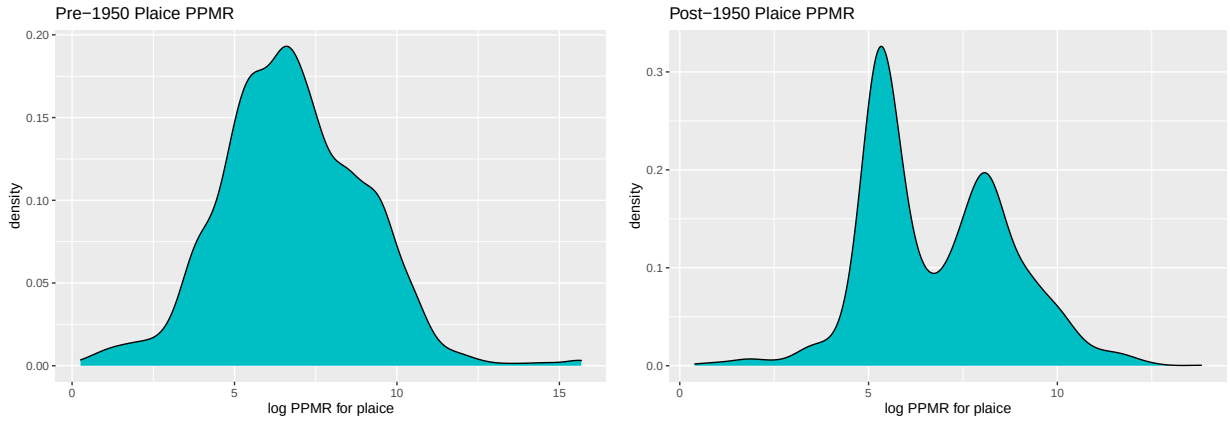


Figure 6: Pre- (left) and post- (right) 1950 log PPMR plots for plaice.

On the other hand, the pre-1950 plaice shows a very different pattern as shown in the left hand side of Figure 6 which seems almost Normal. Consequently, we will try to fit a Normal and a Gamma curve to the data. For the Normal curve with distribution $N(\mu, \sigma^2)$ we will estimate the parameters by $\hat{\mu}$, $\hat{\sigma}^2$ using the mean and the standard deviation of the data giving $\hat{\mu} = 6.7906517$ and $\hat{\sigma}^2 = 4.8090492$. Using the method of moments estimator, for the Gamma distribution we obtain estimates $\hat{\alpha} = 9.5887873$ and $\hat{\beta} = 1.412057$.

Figure 7 shows the fitted Normal distribution in green and the fitted Gamma distribution in red. Upon observation, both are good estimates to the observed data and this is reiterated in the log-likelihoods of -4743.2326851 for the Gamma distribution which corresponds to -2.2673196 per observation and -4610.6617861 for the Normal distribution which corresponds to -2.2039492 per observation. Since the log-likelihood with the Normal distribution is higher, we can consider that it fits the data marginally better than the Gamma distribution.

6.3 Cod

Upon observation, the cod data seem relatively close to a Gamma curve and so we fit such a curve to the data as seen in Figure 8. Using the method of moments, the estimated parameters are $\hat{\alpha} = 11.320626$ and $\hat{\beta} = 1.8456779$.

6.4 Herring

Herring seem to present a greater challenge than other species to fit a curve to due to seeming trimodality. However, we recall that herring suffered a collapse in 1975. Wherefore, we shall plot pre- and post-1975 log PPMR curves separately.

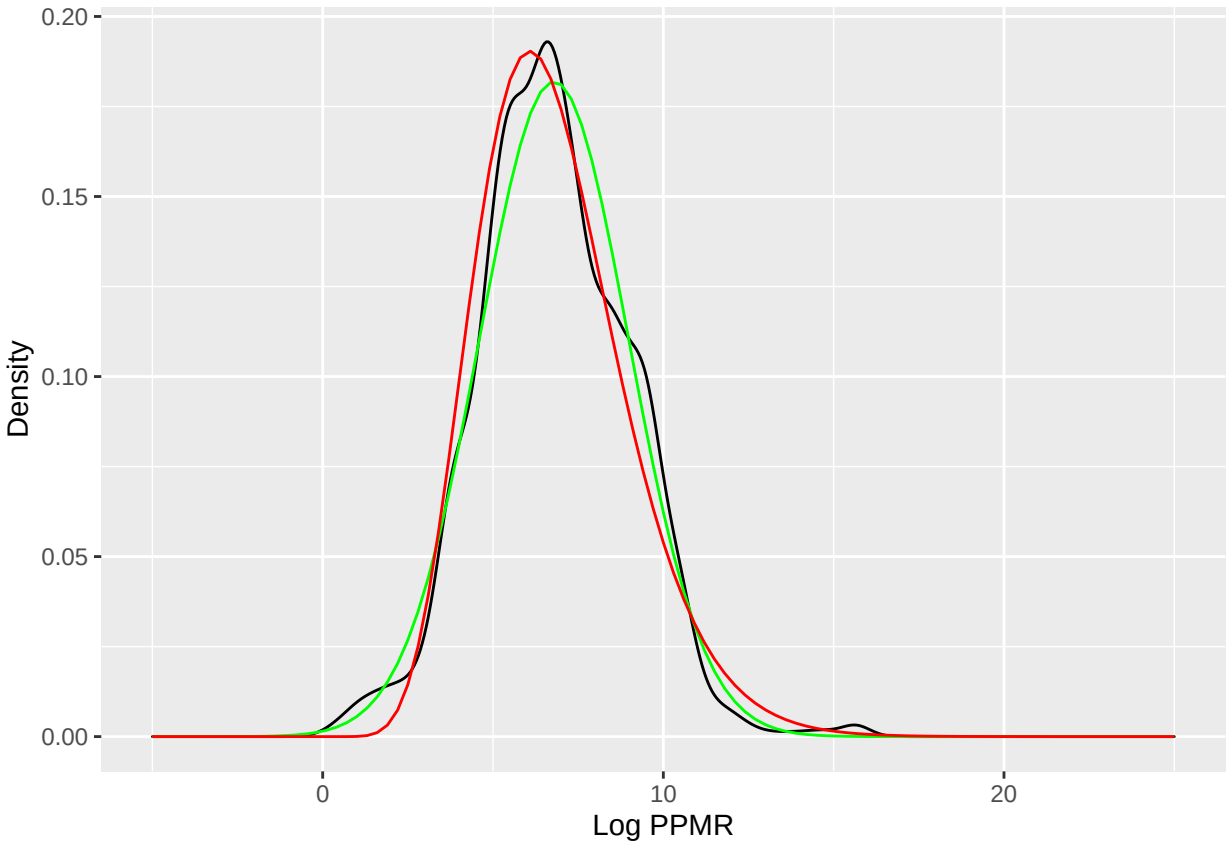


Figure 7: Pre-1950 Plaice log PPMR with fitted Normal (green) and Gamma (red) curves.

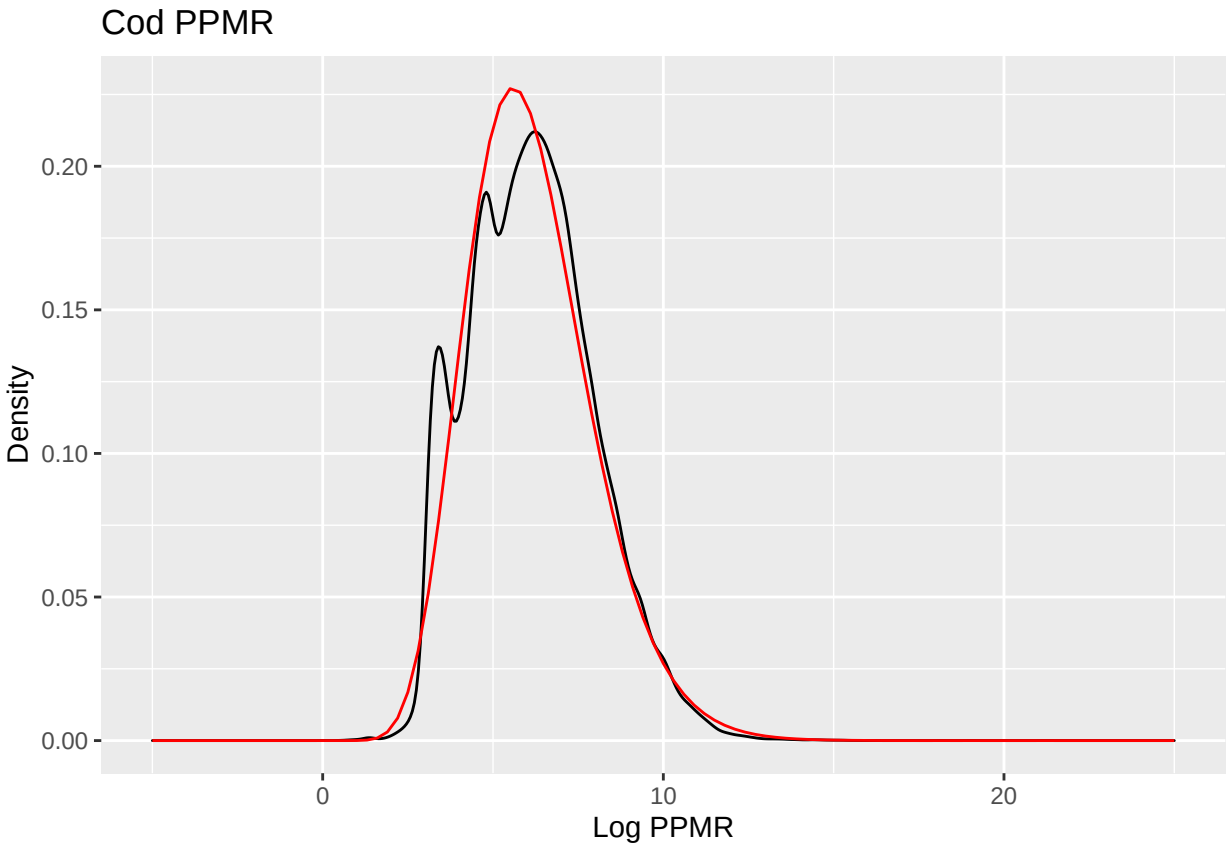


Figure 8: Cod log PPMR with fitted Gamma (red) curve.

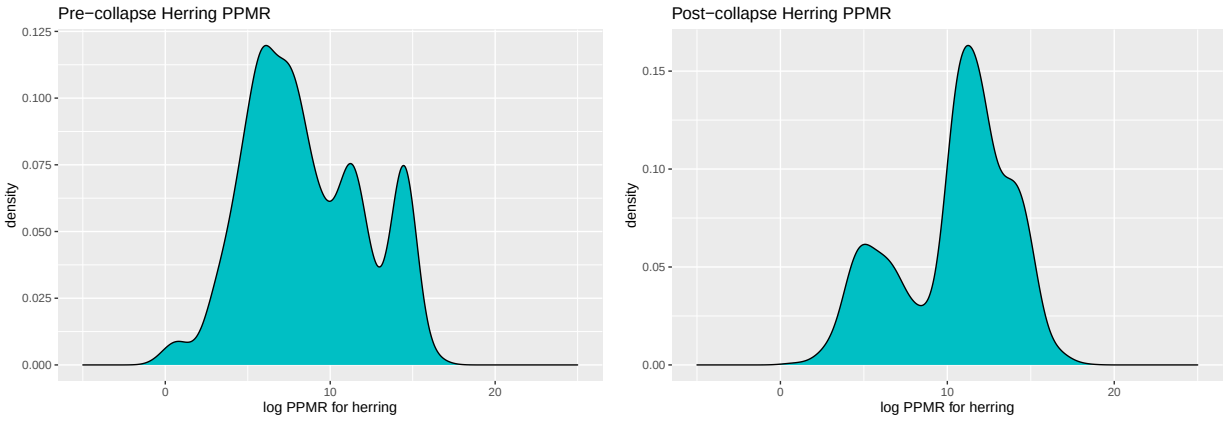


Figure 9: Pre- (left) and post- (right) collapse log PPMR plots for herring.

Clearly, neither of the curves in Figure 9 display features typical of exponential family distributions. Therefore, it is determined that is is inappropriate to fit an exponential family distribution to the herring data.

6.5 Haddock

We may consider from Figure 4 that the log PPMR distribution of haddock has a flatter peak than that of, for example, the Gamma distribution.

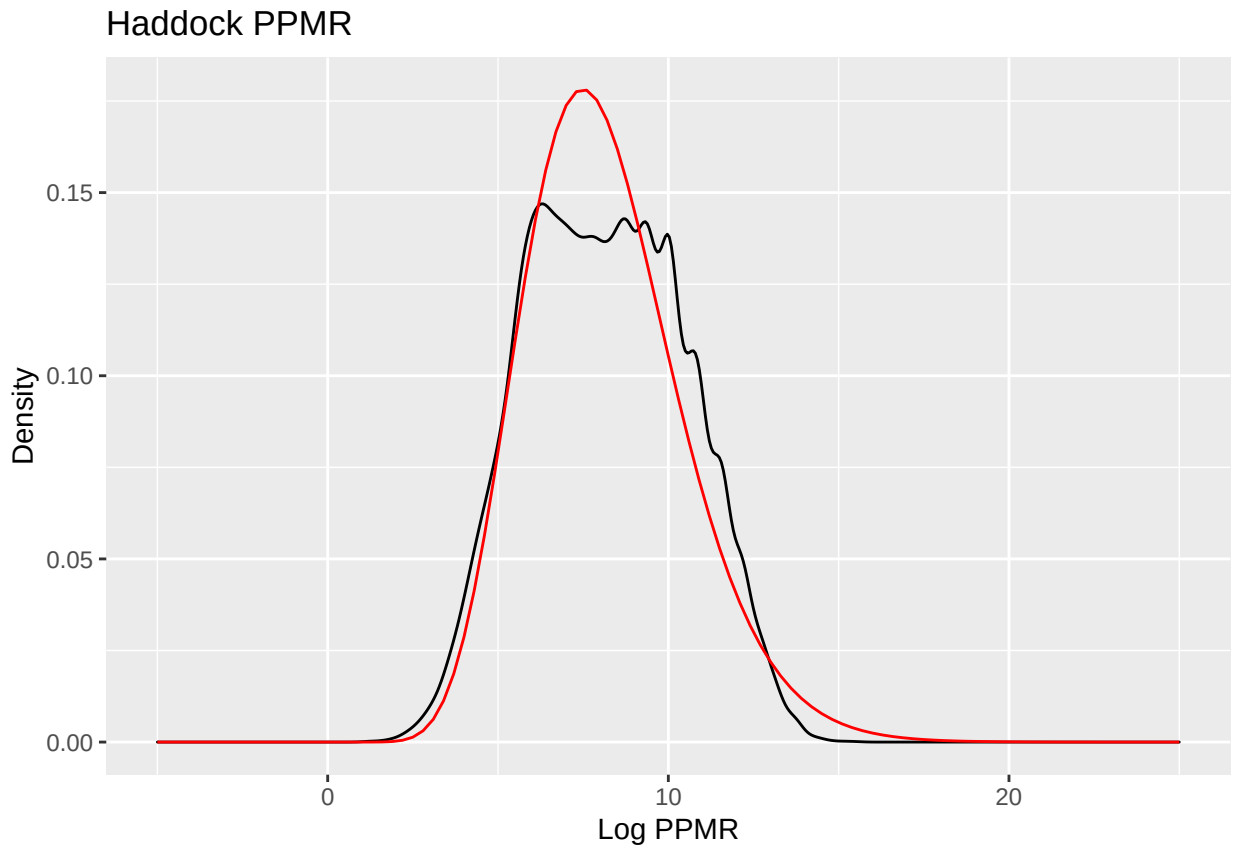


Figure 10: Log PPMR for haddock with fitted Gamma curve.

Nonetheless, we fit a Gamma curve to the data as shown in Figure 10 This is not an unreasonable fit but it is clear that the Gamma distribution has a higher peak. This gives parameter estimates $\hat{\alpha} = 12.3485912$ and $\hat{\beta} = 1.5159218$.

We may consider the geographic distribution of haddock. To do this, we plot observations' latitude and longitude (as a rudimentary map) and note that Greenwich is at (51.5, 0.0) as seen in Figure 11. We note there is a large cluster in the bottom left of the plot which corresponds with the North Sea and seas around the British Isles. However, we also see certain observations relatively far to the north and east which correspond to the Barents Sea which is clearly a much colder environment and so the type of prey may be significantly different there. Consequently, we may choose to focus on the observations near the British Isles. However, this consists of 50184 observations of the 54632. We may however plot the log PPMR with this subset of data as seen in Figure 12 but notice that this does not fit the data much better than Figure 10 and so we determine that the whole haddock dataset is appropriate for use.

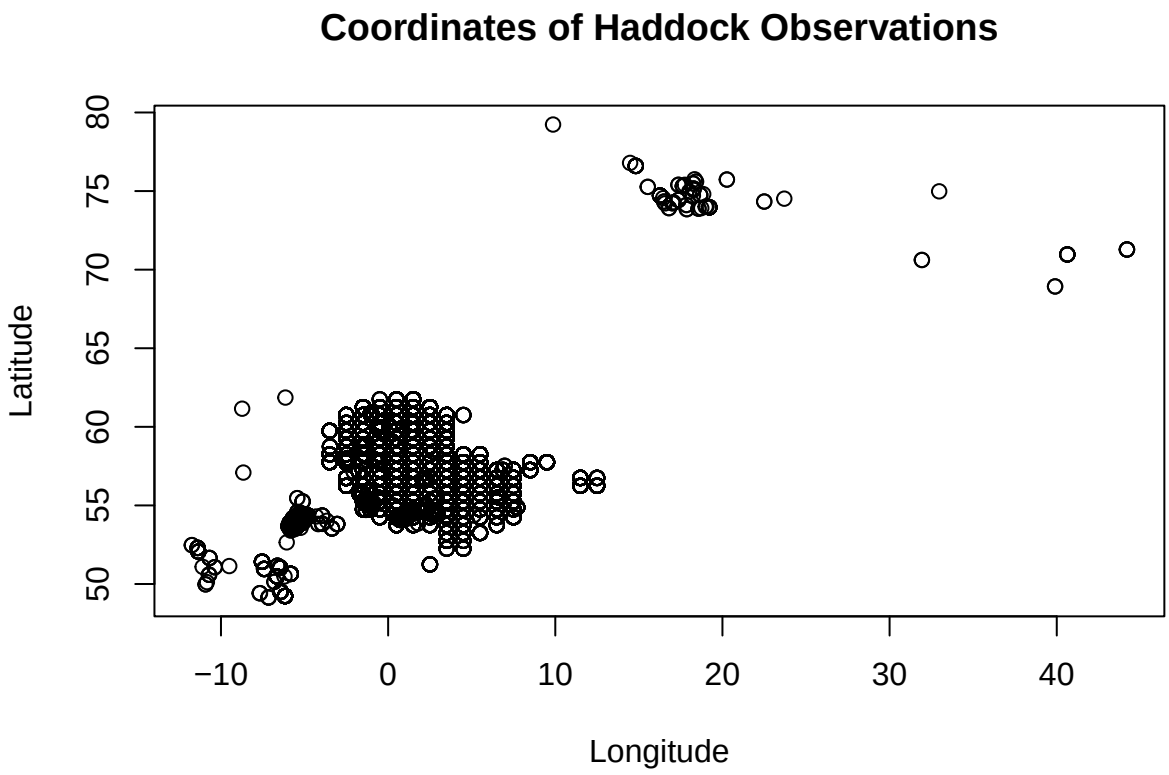


Figure 11: Latitude and longitude of haddock observations.

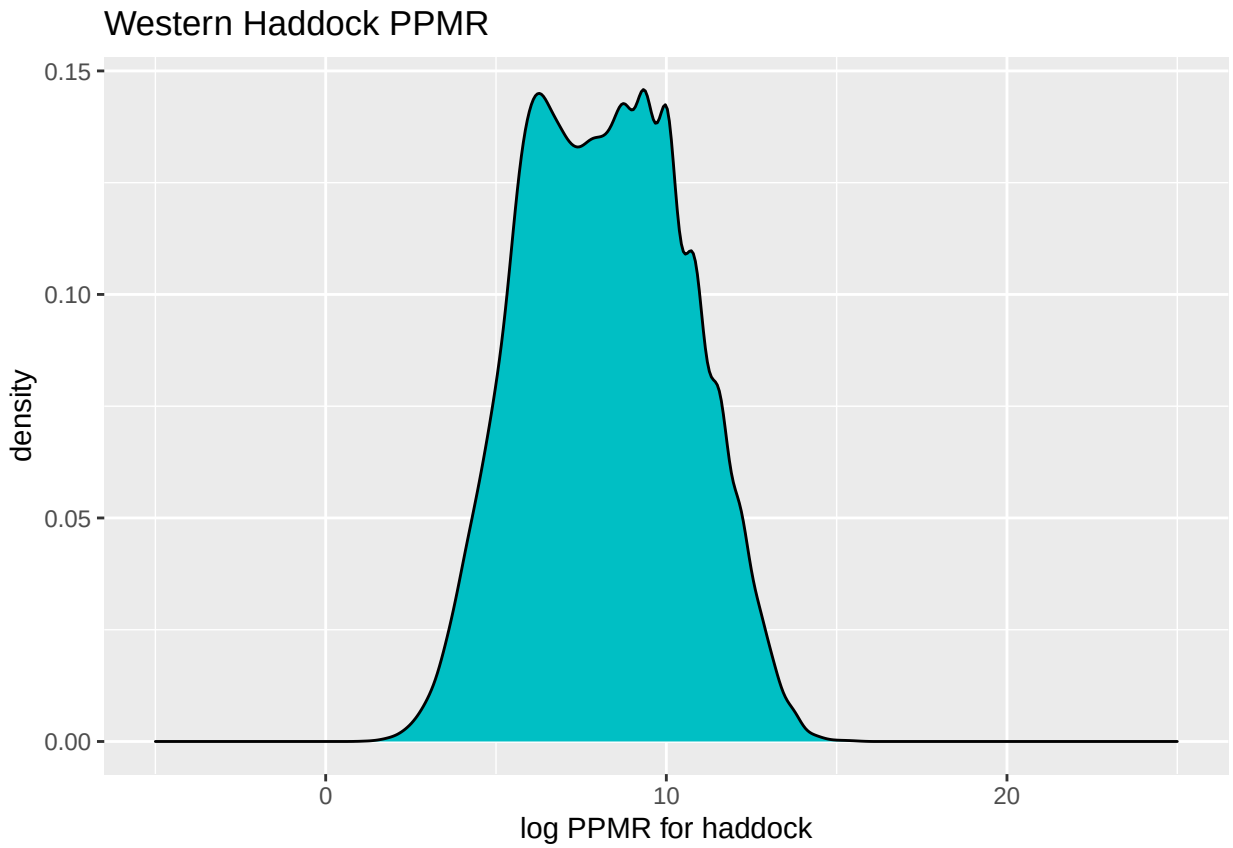


Figure 12: Log PPMR for haddock around the British Isles.

6.6 Sole

The sole data show a similar pattern to the haddock data in having a relatively flat peak, but this is compounded by its bimodality.

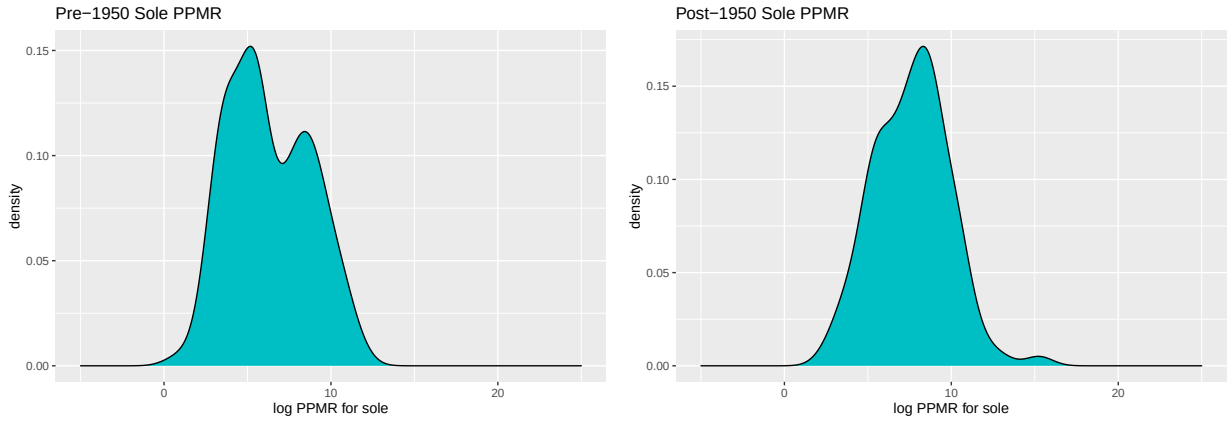


Figure 13: Pre- (left) and post- (right) 1950 log PPMR plots for sole.

We therefore plot pre- and post-1950 plots to ascertain whether the bimodality is a result of long-term changes in the sea as is seen in Figure 13.

Now we must be careful since there are only 219 observations for sole so statistical inference is limited by the small sample size— however we may obtain estimates $\hat{\alpha}$ and $\hat{\beta}$ which will be useful in summarising our results.

7 Summary

Table 1: Summary of parameters for gamma estimate for different species.

Species	Refinement	Condition	$\hat{\alpha}$	$\hat{\beta}$	$\ell(\hat{\alpha},\hat{\beta})/n$
Cod	None	N/A	11.3206259913069	1.84567791145642	-2.00271915712598
Haddock	None	N/A	12.3485911772729	1.51592184374743	-2.27208437568584
Herring	Not Appropriate	N/A	N/A	N/A	N/A
Sole	None	N/A	7.97042594153342	1.15695170306444	-2.32658264506921
Plaice	Time	Pre-1950	9.58878730654174	1.41205700832358	-2.26731963914517
Whiting	None	N/A	6.82801967162711	1.24541171045663	-2.08335735148811

As shown in the Data Results section, the log PPMR of different species varies considerably. However, it seems that fitting a Gamma distribution to the observed log predator-prey mass ratio is appropriate for five out of six studied species.

Table 1 gives a summary of the parameters estimated for a Gamma estimate for each of the species that we have observed as well as the log likelihood divided by the number of observations.

Although there is some variation in $\hat{\alpha}$ and $\hat{\beta}$ from species to species, in general $\hat{\alpha}$ is in the region of 9.6 and $\hat{\beta}$ is in the region of 1.44. This means that it would be reasonable to estimate the observed realised PPMR by a Gamma distribution with these parameters for five of the species we have studied. Using this result and previous conclusions about the link between the observed and preferred PPMR, it would then be possible to use an analytic expression for the feeding kernel in size spectrum models.

This shows that the five species of fish to which it was possible to estimate parameters for showed remarkable regularity in their prey preference. This may be partly due to the fact that these are generally physiologically similar creatures for which it is worthwhile eating prey e^5 to e^{15} times smaller than themselves.

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Appendix

```
knitr::opts_chunk$set(echo = FALSE, message = F, warning = F)
library(dplyr)
library(tidyverse)
library(bbmle)
library(mizer)
library(Hmisc)
library(knitr)
plot(density.default(10, bw=2, kernel = "rectangular"),
     main = "Rectangular Kernel", xlab = "Bandwidth = 2")
plot(density.default(10, bw=2, kernel = "gaussian"),
     main = "Gaussian Kernel", xlab = "Bandwidth = 2")
normSample <- rnorm(100,50,10); normSampDF <- tibble(val = normSample)
ggplot(data = normSampDF) +
  geom_point(aes(val, 0)) +
  xlab("x") +
  theme(axis.title.y = element_blank(),
        axis.text.y = element_blank(), axis.ticks.y = element_blank())
ggplot(data = normSampDF) +
  xlim(10,90) + geom_point(aes(val, 0)) + geom_density(aes(val), kernel = "rectangular") +
  geom_function(fun = dnorm, args = list(mean = 50, sd = 10), linetype = "dashed") +
  labs(y = "Density")
ggplot(data = normSampDF) +
  xlim(10,90) + geom_point(aes(val, 0)) + geom_density(aes(val)) +
  geom_function(fun = dnorm, args = list(mean = 50, sd = 10), linetype = "dashed") +
  labs(y = "Density")
load("data/stomach_dataset.Rdata")

df <- stom_df %>%
  transmute(Species = pred_species,
            latitude,
            longitude,
            year,
            indwprey = prey_ind_weight_g,
            totwprey = prey_weight_g,
            ppmr,
            wpredator = pred_weight_g,
            Nprey = prey_count / n_stomachs)

df = df %>%
  filter(df$indwprey != 0)

df = df %>%
  filter(df$indwprey != Inf)

df = df %>%
  mutate(pmbr = wpredator/totwprey, lppmr = log(ppmr), lpmbr = log(pmbr))

choicedf = df %>%
  filter(Species == "Gadus morhua" |
           Species == "Melanogrammus aeglefinus" |
           Species == "Clupea harengus" |
```

```
           Species == "Pleuronectes platessa" |
           Species == "Solea solea" |
           Species == "Merlangius merlangus")

choicedf = choicedf %>%
  mutate(Species = replace(Species, Species == "Gadus morhua", "Cod")) %>%
  mutate(Species = replace(Species,
                       Species == "Melanogrammus aeglefinus", "Haddock")) %>%
  mutate(Species = replace(Species, Species == "Clupea harengus", "Herring")) %>%
  mutate(Species = replace(Species, Species == "Pleuronectes platessa", "Plaice")) %>%
  mutate(Species = replace(Species, Species == "Solea solea", "Sole")) %>%
  mutate(Species = replace(Species, Species == "Merlangius merlangus", "Whiting"))


codDF <- filter(df, df$Species == "Gadus morhua")
haddockDF <- filter(df, df$Species == "Melanogrammus aeglefinus")
herringDF <- filter(df, df$Species == "Clupea harengus")
plaiceDF <- filter(df, df$Species == "Pleuronectes platessa")
soleDF <- filter(df, df$Species == "Solea solea")
whitingDF <- filter(df, df$Species == "Merlangius merlangus")
allgg <- choicedf %>%
  ggplot(aes(lppmr)) +
    geom_density(aes(lppmr), fill = "#00BFC4") +
    labs(title = "PPMR for Species", x = "log PPMR") +
    facet_wrap(vars(Species), nrow = 3)

allgg
whitingPPMRAlphaMOM = mean(whitingDF$lppmr)^2/(var(whitingDF$lppmr))
whitingPPMRBetaMOM = mean(whitingDF$lppmr)/(var(whitingDF$lppmr))
ggplot(whitingDF, aes(x = lppmr), adjust = 1.5, xlab = "Log PPMR") +
  geom_density() +
  stat_function(fun = dgamma,
      args = list(shape = whitingPPMRAlphaMOM, scale = 1/whitingPPMRBetaMOM),
      col = "red") +
  xlim(0,15) +
  labs(title = "Whiting PPMR", x = "Log PPMR", y = "density")
whitingLogLik = 0.0
for (i in 1:nrow(whitingDF)) {
  whitingLogLik = whitingLogLik +
    log(dgamma(whitingDF$lppmr[i] , shape = whitingPPMRAlphaMOM,
               scale = 1/whitingPPMRBetaMOM))
}
modernPlaice <- plaiceDF %>%
  filter(year >= 1950)
oldPlaice <- plaiceDF %>%
  filter(year < 1950)
modPlaicePlot <- ggplot(modernPlaice, aes(x = lppmr)) +
  geom_density(fill = "#00BFC4") +
  labs(title = "Post-1950 Plaice PPMR", x = "log PPMR for plaice")
oldPlaicePlot <- ggplot(oldPlaice, aes(x = lppmr)) +
  geom_density(fill = "#00BFC4") +
  labs(title = "Pre-1950 Plaice PPMR", x = "log PPMR for plaice")
oldPlaicePlot
```

```
modPlaicePlot
meanOldPlaice <- mean(oldPlaice$lppmr)
sdOldPlaice <- sd(oldPlaice$lppmr)

oldPlaiceAlpha <- mean(oldPlaice$lppmr)^2/(var(oldPlaice$lppmr))
oldPlaiceBeta <- mean(oldPlaice$lppmr)/(var(oldPlaice$lppmr))
ggplot(oldPlaice, aes(x = lppmr), adjust = 1.5, xlab = "Log PPMR") +
  geom_density() +
  stat_function(fun = dnorm,
      args = list(mean = meanOldPlaice, sd = sdOldPlaice), col = "green") +
  stat_function(fun = dgamma,
        args = list(shape = oldPlaiceAlpha, scale = 1/oldPlaiceBeta), col = "red") +
  xlim(-5,25) +
  xlab("Log PPMR") +
  ylab("Density") +
  labs(main = "Pre-1950 Plaice PPMR")
oldPlaiceGammaLogLik = 0.0
for (i in 1:nrow(oldPlaice)) {
  oldPlaiceGammaLogLik = oldPlaiceGammaLogLik +
    log(dgamma(oldPlaice$lppmr[i] , shape = oldPlaiceAlpha,
               scale = 1/oldPlaiceBeta))
}

oldPlaiceNormalLogLik = 0.0
for (i in 1:nrow(oldPlaice)) {
  oldPlaiceNormalLogLik = oldPlaiceNormalLogLik +
    log(dnorm(oldPlaice$lppmr[i] , mean = meanOldPlaice, sd = sdOldPlaice))
}
codAlpha <- mean(codDF$lppmr)^2/(var(codDF$lppmr))
codBeta <- mean(codDF$lppmr)/(var(codDF$lppmr))
ggplot(codDF, aes(x = lppmr), adjust = 1.5, xlab = "Log PPMR") +
  geom_density() +
  stat_function(fun = dgamma,
        args = list(shape = codAlpha, scale = 1/codBeta), col = "red") +
  xlim(-5,25) +
  labs(title = "Cod PPMR") +
  xlab("Log PPMR") +
  ylab("Density")
codLogLik = 0.0
for (i in 1:nrow(codDF)) {
  codLogLik = codLogLik +
    log(dgamma(codDF$lppmr[i] , shape = codAlpha,
               scale = 1/codBeta))
}
preHerring <- herringDF %>%
  filter(year <= 1975)
postHerring <- herringDF %>%
  filter(year > 1975)
herringPreCollapsegg <- ggplot(preHerring, aes(x = lppmr)) +
  geom_density(fill = "#00BFC4") +
  labs(title = "Pre-collapse Herring PPMR", x = "log PPMR for herring") +
  xlim(-5,25)
herringPostCollapsegg <- ggplot(postHerring, aes(x = lppmr)) +
```

```
  geom_density(fill = "#00BFC4") +
  labs(title = "Post-collapse Herring PPMR", x = "log PPMR for herring") +
  xlim(-5,25)

herringPreCollapsegg
herringPostCollapsegg
haddockAlpha <- mean(haddockDF$lppmr)^2/(var(haddockDF$lppmr))
haddockBeta <- mean(haddockDF$lppmr)/(var(haddockDF$lppmr))
ggplot(haddockDF, aes(x = lppmr), adjust = 1.5) +
  geom_density() +
  stat_function(fun = dgamma,
        args = list(shape = haddockAlpha, scale = 1/haddockBeta), col = "red") +
  xlim(-5,25) +
  labs(title = "Haddock PPMR", x = "Log PPMR", y = "Density")
haddockLogLik = 0.0
for (i in 1:nrow(haddockDF)) {
  haddockLogLik = haddockLogLik +
    log(dgamma(haddockDF$lppmr[i] , shape = haddockAlpha,
               scale = 1/haddockBeta))
}
plot(x = haddockDF$longitude, y = haddockDF$latitude,
     xlab = "Longitude", ylab = "Latitude", main = "Coordinates of Haddock Observations")
westernHaddock = haddockDF %>%
  filter(longitude < 20) %>%
  filter(latitude < 65)
ggplot(westernHaddock, aes(x = lppmr)) +
  geom_density(fill = "#00BFC4") +
  labs(title = "Western Haddock PPMR", x = "log PPMR for haddock") +
  xlim(-5,25)
oldSole <- soleDF %>%
  filter(year < 1950)

modernSole <- soleDF %>%
  filter(year >= 1950)
soleAlpha <- mean(soleDF$lppmr)^2/(var(soleDF$lppmr))
soleBeta <- mean(soleDF$lppmr)/(var(soleDF$lppmr))
oldSolegg <- ggplot(oldSole, aes(x = lppmr)) +
  geom_density(fill = "#00BFC4") +
  labs(title = "Pre-1950 Sole PPMR", x = "log PPMR for sole") +
  xlim(-5,25)
modSolegg <- ggplot(modernSole, aes(x = lppmr)) +
  geom_density(fill = "#00BFC4") +
  labs(title = "Post-1950 Sole PPMR", x = "log PPMR for sole") +
  xlim(-5,25)

oldSolegg
modSolegg
soleLogLik = 0.0
for (i in 1:nrow(soleDF)) {
  soleLogLik = soleLogLik +
    log(dgamma(soleDF$lppmr[i] , shape = soleAlpha,
               scale = 1/soleBeta))
}
```

```
speciesNames = c("Cod", "Haddock", "Herring", "Sole", "Plaice", "Whiting")
refinements = c("None", "None", "Not Appropriate", "None", "Time", "None")
conditionRefines = c("N/A", "N/A", "N/A", "N/A", "Pre-1950", "N/A")
alphaEstimates = c(codAlpha, haddockAlpha, "N/A",
    soleAlpha, oldPlaiceAlpha, whitingPPMRAlphaMOM)
betaEstimates = c(codBeta, haddockBeta, "N/A",
    soleBeta, oldPlaiceBeta, whitingPPMRBetaMOM)
logLiks = c(codLogLik/nrow(codDF), haddockLogLik/nrow(haddockDF), "N/A",
            soleLogLik/nrow(soleDF),
            oldPlaiceGammaLogLik/nrow(oldPlaice), whitingLogLik/nrow(whitingDF))

gammaResultsTable = cbind(speciesNames, refinements,
              conditionRefines, alphaEstimates, betaEstimates, logLiks)
gammaResultsMat = matrix(gammaResultsTable, 6, 6)

kable(gammaResultsMat, col.names = c("Species", "Refinement",
      "Condition", "$\\hat{\\alpha}$", "$\\hat{\\beta}$",
      "$\\ell(\\hat{\\alpha},\\hat{\\beta})/n$"),
      digits = 2, escape = F,
      caption = "\\label{gammSumTab}Summary of parameters for
      gamma estimate for different species.")
```